

# Reproducibility

Reproducible computational research requires version-pinned tools, fixed inputs, and documented outputs [peng2011reproducible]. COGANT contributes the **graph-generation** slice of that story: identical source trees and identical pipeline configuration should yield identical exported bundles modulo declared nondeterminism (for example optional neural embeddings if enabled).

## Version pinning

Pin:

- ▶ The **COGANT** version (`pyproject.toml` / package `__version__`) and Git commit when installing from source.
- ▶ The **Python** minor version.
- ▶ **Optional** Rust toolchain commit when building native extensions from `../cogant/rust/`.

## Artifact layout

Pipeline output typically includes JSON for intermediate IRs, **Generalized Notation Notation (GNN)** bundles (canonical `model.gnn.md` plus companion JSON and interop artifacts),